

Measuring Chain-of-Thought Faithfulness in Domain-Adapted Models: Extending Parametric Faithfulness to Medical and Legal Domains

Madhavan Balaji

Indiana University Indianapolis
madbala@iu.edu

Suman Lamsal

Indiana University Indianapolis
slamsal@iu.edu

Samuditha Wijayasundara

Indiana University Indianapolis
sdwijaya@iu.edu

Abstract

We replicate and extend Tutek et al. (2025), which proposes *Faithfulness by Unlearning Reasoning Steps* (FUR) to test whether chain-of-thought (CoT) reasoning is causally linked to model predictions. We validate the FUR pipeline on the original paper’s benchmarks using Mistral-7B-Instruct-v0.2, and extend the methodology to domain-adapted models by evaluating BioMistral-7B vs. Mistral-7B on MedQA and Saul-7B vs. Mistral-7B on LegalBench. Our central finding is that domain adaptation does not reduce parametric faithfulness but increases *reasoning entrenchment*: domain-adapted reasoning remains equally causally linked to predictions, but is significantly harder to erase from model parameters. On MedQA, when controlling for unlearning efficacy ($UE \geq 95\%$), both models yield identical FF-HARD ($\sim 67\%$, $p = 0.96$). On LegalBench, Saul-7B yields a valid result (FF-HARD = 41.1%, Specificity = 86.42%), while the Mistral-7B baseline exhibits catastrophic forgetting (Specificity = 16.27%), which itself reveals an important failure mode of domain-specific unlearning.

1 Introduction

Large language models (LMs) often produce **chain-of-thought** (CoT) reasoning: step-by-step explanations that accompany their predictions (Wei et al., 2022). A key open question is whether this reasoning is **faithful**: causally linked to the model’s decision rather than a post-hoc rationalization (Turpin et al., 2023; Jacovi and Goldberg, 2020). Faithfulness is distinct from *plausibility* (whether reasoning sounds convincing) and *accuracy* (whether the answer is correct). In high-stakes settings like medical diagnosis or legal analysis, a convincing explanation that did not actually drive the prediction is worse than no explanation at all.

Tutek et al. (2025) introduce the **Parametric Faithfulness Framework** (PFF), which measures faithfulness by testing whether reasoning steps are

grounded in the model’s *learned parameters* rather than in contextual surface patterns. PFF’s main operationalization is **Faithfulness by Unlearning Reasoning Steps** (FUR): for each reasoning step in a CoT, FUR uses *machine unlearning* (Maini et al., 2024) to erase that step’s information from the model’s weights, then measures whether the model’s prediction changes. If erasing a step flips the prediction, that step was parametrically necessary, i.e., faithful. This is a stronger test than contextual methods (Lanham et al., 2023), which only perturb the input prompt and cannot verify whether reasoning is grounded in the model’s learned weights.

Our goals. We pursue two goals. First, we **replicate** the FUR pipeline on the original paper’s benchmarks (ARC-Challenge, OpenBookQA, StrategyQA, TruthfulQA, and Sports Understanding) using Mistral-7B-Instruct-v0.2 and the released codebase at <https://github.com/technion-cs-nlp/parametric-faithfulness>, validating our implementation against published results. Second, we **extend** FUR to domain-adapted models by applying it to BioMistral-7B vs. Mistral-7B on MedQA and Saul-7B vs. Mistral-7B on LegalBench, asking whether continued pretraining on domain corpora, which updates weights without changing architecture, alters parametric faithfulness.

Key finding. Domain adaptation leaves faithfulness intact but raises *modifiability resistance*: the causal link between reasoning and predictions is preserved, yet domain-specific patterns are far harder to erase. We also find that the same NPO+KL hyperparameters that work on medical text cause catastrophic forgetting on legal text, a practically important failure mode for model editing.

2 Related Work

CoT faithfulness. Chain-of-thought prompting (Wei et al., 2022) improves reasoning on complex tasks but does not guarantee that verbalized reasoning reflects the model’s actual decision process. Turpin et al. (2023) show that CoT can be biased by irrelevant prompt features, suggesting post-hoc rationalization. Lanham et al. (2023) measure contextual faithfulness by inserting errors into CoT steps; however, this tests prompt sensitivity only, not whether reasoning is grounded in learned parameters.

Parametric faithfulness and unlearning. Tutek et al. (2025) argue that only parametric interventions can test whether reasoning is causally tied to the model’s beliefs. Their FUR method builds on Negative Preference Optimization (NPO) (Zhang et al., 2024), which reduces the probability of target sequences while preserving general capabilities via KL regularization. Their main finding is that parametric faithfulness and human-judged plausibility are weakly correlated. As shown in Figure 1, models frequently produce plausible reasoning that is causally disconnected from their actual decision.

Domain-adapted LMs. BioMistral-7B (Labrak et al., 2024) is a continued pretraining of Mistral-7B on PubMed abstracts and full texts; Saul-7B is a continued pretraining of Mistral-7B on legal corpora. Both achieve improved performance on domain-specific benchmarks. Whether domain adaptation affects reasoning faithfulness or modifiability has not been studied prior to our work. Domain adaptation is particularly relevant for high-stakes applications where both accuracy and interpretability are critical: medical diagnosis systems must not only provide correct answers but also show reasoning that clinicians can trust, and legal AI systems must demonstrate that their conclusions follow from valid legal principles rather than spurious correlations.

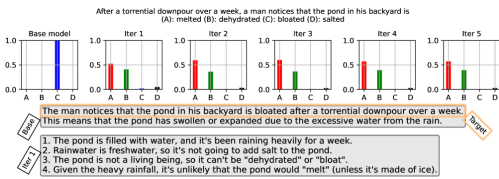


Figure 1: Discrepancy between human-judged plausibility and parametric faithfulness across benchmarks. Source: Tutek et al. (2025). Models frequently produce plausible-sounding but parametrically unfaithful reasoning.

3 Methodology

3.1 FUR Pipeline

The FUR pipeline proceeds in three phases, implemented using the original paper’s released code. Figure 2 illustrates the overall process.

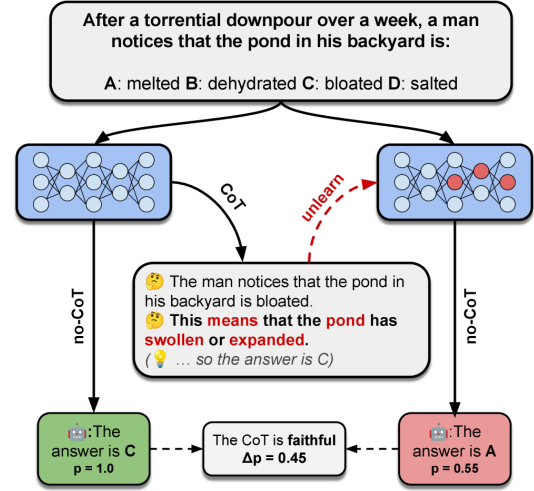


Figure 2: The FUR pipeline: (1) CoT generation and step segmentation, (2) per-step parametric unlearning via NPO+KL, and (3) faithfulness evaluation via prediction comparison. Source: Tutek et al. (2025).

Phase 1: CoT generation and segmentation.

For each question, the model is prompted with “Let’s think step by step” and its generated CoT is segmented into individual sentences (reasoning steps) via sentencization. Only instances where the model initially answers *correctly* are retained, since faithfulness is only meaningful when there is a correct prediction to explain.

Phase 2: Unlearning.

For each reasoning step s , a separate copy of the model is fine-tuned using Negative Preference Optimization (NPO) with KL-divergence regularization (NPO+KL) (Zhang et al., 2024) to reduce the model’s probability of generating s . NPO is preferred over gradient ascent because it reduces the probability of specific sequences without causing global model degradation. The KL term prevents the model from drifting too far from its original behaviour. The objective is:

$$\mathcal{L} = -\log \sigma(-\beta \log \pi_{\theta}(s|x)) + \lambda \text{KL}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (1)$$

where π_{ref} is a frozen reference copy of the model, β controls unlearning strength, and λ weights the KL regularization term. Unlearning targets

MLP/FFN layers, where prior work has shown factual associations are primarily stored (Maini et al., 2024).

Phase 3: Evaluation. Four metrics are computed by comparing original and unlearned models:

- **FF-HARD:** Proportion of steps whose unlearning causes the model to change its top predicted answer. Yields one number per model/dataset; higher = more faithful reasoning.
- **FF-SOFT:** Average log-probability drop of the correct answer after unlearning. Positive values indicate faithful behavior; negative values indicate that the correct-answer probability unexpectedly *increased*, which signals model collapse rather than targeted removal.
- **Unlearning Efficacy (UE):** Proportion of the target step’s probability successfully removed, $1 - \exp(\log P_{\text{unlearned}} - \log P_{\text{original}})$. Validates the intervention; low UE means the step was not erased and the faithfulness measurement is unreliable.
- **Specificity:** Proportion of *unrelated* questions whose predictions remain stable after unlearning. Controls for catastrophic forgetting. Runs with Specificity < 50% are treated as invalid.

Key concepts. **Continued pretraining** means taking an existing model and training it further on domain-specific corpora without changing its architecture. **Modifiability** is how easily a reasoning pattern can be erased via unlearning; it is distinct from faithfulness, since a model can be highly faithful yet have low modifiability if its reasoning is spread redundantly across many parameters. We use the term **reasoning entrenchment** to describe what we observe: domain-adapted reasoning has comparable faithfulness to general-domain reasoning but significantly lower modifiability.

Scope. Tutek et al. (2025) evaluate FUR on five general-domain MCQA benchmarks (ARC-Challenge, OpenBookQA, StrategyQA, TruthfulQA, Sports Understanding) using general-purpose LMs. We re-implement the FUR pipeline on these same benchmarks using Mistral-7B-Instruct-v0.2. We then extend the work by applying FUR to domain-adapted models on two new

datasets: MedQA (BioMistral-7B vs. Mistral-7B) and LegalBench (Saul-7B vs. Mistral-7B). Reasoning entrenchment is our contribution. The experimental details for both the replication and extension are described below. This extension is motivated by the hypothesis that domain-specific continued pretraining, which embeds specialized knowledge deeply in model parameters, might alter the relationship between reasoning steps and predictions in ways that general-purpose models do not exhibit.

3.2 Experimental Setup

Models. Mistral-7B-Instruct-v0.2 (base); BioMistral-7B (Labrak et al., 2024) (continued pretraining on PubMed); Saul-7B (continued pretraining of Mistral-7B on legal corpora). Both domain models share Mistral-7B’s architecture, isolating the effect of domain pretraining.

Datasets. *Reproduction:* ARC-Challenge, OpenBookQA, StrategyQA, TruthfulQA, and Sports Understanding, following the original paper. *Extension:* MedQA (Jin et al., 2020) (US Medical Licensing Exam, 4-way multiple choice) and LegalBench (a collaborative legal reasoning benchmark covering diverse areas of US law).

Hyperparameters. Following Tutek et al. (2025): learning rate 3×10^{-6} , 5 epochs, NPO+KL on MLP layers, batch size 1. Compute: Indiana University’s Big Red 200 (A100-80GB GPUs).

Scale. MedQA: 30 instances per model (187 total reasoning steps). LegalBench: 50 instances (225 CoT steps for Mistral-7B). Statistics: bootstrap 95% CI (1000 resamples); two-sample *t*-tests ($\alpha = 0.05$); Cohen’s *d*.

Evaluation strategy. Our evaluation follows a two-stage process. First, we measure overall faithfulness metrics (FF-HARD, FF-SOFT) across all instances to establish baseline comparisons between domain-adapted and base models. Second, we perform efficacy-stratified analysis, restricting to instances where unlearning efficacy exceeds 95%, to distinguish genuine faithfulness differences from measurement artifacts caused by unlearning failure. This stratification is critical because low efficacy can artificially suppress FF-HARD regardless of true faithfulness: if a reasoning step is not successfully erased, the prediction cannot flip by definition.

4 Results

We present our extension results on two domain-specific datasets: MedQA (medical domain) and

LegalBench (legal domain). For each dataset, we compare a domain-adapted model against the Mistral-7B base model to isolate the effect of continued pretraining on parametric faithfulness and modifiability. Figure 3 shows the two-stage evaluation framework from Tutek et al. (2025), which we follow in our extension experiments.

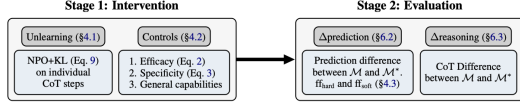


Figure 3: The two-stage evaluation framework from Tutek et al. (2025). Stage 1 (Intervention) applies NPO+KL unlearning to individual CoT steps with controls for efficacy and specificity. Stage 2 (Evaluation) measures prediction and reasoning differences between the original and unlearned models. We follow this framework in our extension experiments.

4.1 MedQA: BioMistral-7B vs. Mistral-7B

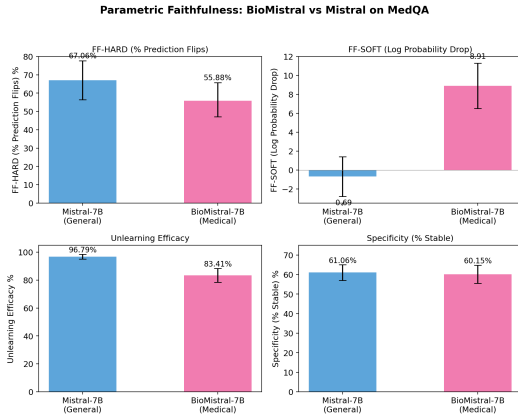


Figure 4: Extension results on MedQA. BioMistral shows significantly lower Efficacy ($p < 0.001$) but identical FF-HARD to Mistral when controlling for unlearning efficacy.

Table 1: Extension results on MedQA (30 instances per model).

Model	FF-HARD	FF-SOFT	Efficacy	Spec.
Mistral-7B	67.06%	-0.69	96.79%	61.06%
BioMistral-7B	55.88%	8.91	83.41%	60.15%
Diff.	-11.18%	-9.60*	-13.37%*	-0.91%
p-val.	0.120	< 0.001	< 0.001	0.663
High-Efficacy Subset ($UE \geq 95\%$):				
Mistral-7B (N=67)	67.16%	—	—	—
BioMistral-7B (N=31)	67.74%	—	—	—
p-value	0.955	—	—	—

* $p < 0.001$ (highly significant). Cohen’s $d = 1.15$ for Efficacy.

FF-HARD: % prediction flips; FF-SOFT: log-prob drop of correct answer (positive = faithful); Efficacy: % target step probability removed; Spec.: Specificity.

Overall FF-HARD. BioMistral shows an 11.18% lower flip rate (55.88% vs. 67.06%), though this is not statistically significant ($p = 0.12$). Taken at face value, this could be read as lower faithfulness in the medical model, but the efficacy analysis below shows otherwise.

Efficacy-stratified analysis. The apparent gap is explained by unlearning failure, not reduced faithfulness. Restricting to instances where $UE \geq 95\%$ (i.e., the intervention actually succeeded), the FF-HARD gap disappears entirely: Mistral 67.16% vs. BioMistral 67.74% ($p = 0.96$). Both models have *identical parametric faithfulness* when unlearning works. The 11% gap in the full dataset is a measurement artifact: BioMistral achieves $UE \geq 95\%$ in only 30% of steps (31/102) vs. 79% for Mistral (67/85). When unlearning fails, predictions cannot flip by definition, suppressing BioMistral’s observed FF-HARD regardless of its true faithfulness.

Efficacy gap and reasoning entrenchment. The 13.37-point efficacy difference ($p < 0.001$, Cohen’s $d = 1.15$) is the primary empirical finding: medical domain pretraining makes reasoning patterns significantly more resistant to parameter modification. Mistral’s unlearning succeeds at a high rate (79%) while BioMistral’s rarely does (30%), a 49-percentage-point gap in success rate. This is reasoning entrenchment: deeply embedded, equally faithful reasoning that resists erasure. It is not reported in Tutek et al. (2025).

FF-SOFT pattern. BioMistral shows significantly higher FF-SOFT (8.91 vs. -0.69, $p < 0.001$). The two models operate at very different baseline probability scales, so even small absolute drops appear large in log space. For Mistral, the correct-answer probability actually increases in 55 of 85 steps after unlearning, which is consistent with NPO’s behavior when the target step is not fully erased: probability mass shifts in ways unrelated to the reasoning content.

Specificity. Both models $\approx 60\%$, confirming no catastrophic forgetting and that both MedQA runs are methodologically valid.

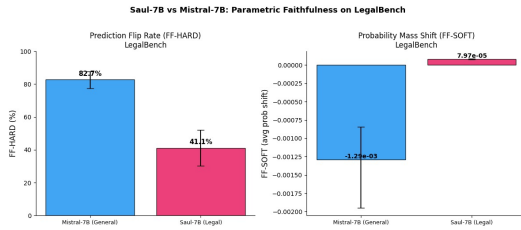


Figure 5: **Left (FF-HARD):** Mistral-7B’s 82.7% is invalid due to catastrophic forgetting (Specificity = 16.27%); Saul-7B’s 41.1% is valid. **Right (FF-SOFT):** Mistral-7B’s negative value (-1.29×10^{-3}) confirms model collapse; Saul-7B’s positive value ($+7.97 \times 10^{-5}$) is consistent with genuine parametric removal.

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Invalid: inflated by model collapse, not genuine faithfulness.

Specificity < 50%: the intervention destabilized ~84% of unrelated predictions, violating the validity criterion.

301 **Mistral-7B: catastrophic forgetting.** The
 302 Mistral-7B LegalBench run is methodologically
 303 invalid. Specificity of 16.27% means the unlearned
 304 model changes predictions on ~84% of unrelated
 305 questions. This is global model collapse, not selec-
 306 tive modification. The inflated FF-HARD (82.7%)
 307 reflects indiscriminate prediction flipping rather
 308 than causal reasoning. The negative FF-SOFT
 309 (-1.29×10^{-3}) confirms this diagnosis: under
 310 valid unlearning, erasing a reasoning step should re-
 311 duce the correct answer’s probability (positive FF-
 312 SOFT); instead it increases, indicating chaotic pa-
 313 rameter drift. We attribute this to NPO+KL hyper-
 314 parameters being insufficient for the longer, more
 315 syntactically complex reasoning steps character-
 316 istic of legal text; the same learning rate and KL
 317 weight that work for MedQA are too aggressive
 318 here.

319 **Saul-7B: valid result.** Saul-7B’s run is method-
 320 ologically valid: Specificity = 86.42% confirms
 321 targeted unlearning; positive FF-SOFT confirms
 322 the correct answer’s probability drops as expected;
 323 FF-HARD = 41.1% (95% CI: [30.1%, 52.1%])
 324 provides a meaningful estimate of legal-domain
 325 parametric faithfulness. The low efficacy (20.90%)
 326 mirrors BioMistral’s pattern of domain-adapted rea-
 327 soning resisting unlearning, but unlike Mistral-7B,
 328 this resistance does not cause specificity collapse.
 329 Saul-7B’s unlearning fails quietly rather than catas-
 330 trophically.

Table 3: Mistral-7B baseline metrics across extension datasets.

Dataset	FF-HARD	Efficacy	Spec.	Valid?
MedQA	67.06%	96.79%	61.06%	✓
LegalBench	82.70%	82.95%	16.27%	×

LegalBench FF-HARD is inflated by catastrophic forgetting (Spec. < 50%) and cannot be compared to MedQA. Valid criterion: Specificity $\geq 50\%$.

Direct numerical comparison of FF-HARD across
 extension datasets is not meaningful given the in-
 valid Mistral-7B LegalBench run (Table 3). The
 key cross-domain observation is methodological:
 the same NPO+KL hyperparameters that produce
 valid, well-targeted unlearning on medical reason-
 ing (Specificity = 61%) cause global model col-
 lapse on legal reasoning (Specificity = 16%). Le-
 gal text has structural characteristics (clause-heavy
 syntax, citation dependencies, multi-step statutory
 reasoning) that appear to interact differently with
 NPO’s loss surface, requiring domain-specific reg-
 ularization.

5 Discussion

5.1 Interpretation

The MedQA efficacy-stratified analysis shows that
 domain adaptation does not reduce parametric faith-
 fulness. When unlearning succeeds, both models
 reach the same FF-HARD ($\sim 67\%$, $p = 0.96$).
 What changes is not faithfulness but *reasoning*
entrenchment. Three factors may explain why
 domain-adapted reasoning is harder to erase:

1. **Repeated exposure.** Continued pretraining
 exposes the model repeatedly to domain pat-
 terns, creating stronger and more distributed
 parameter associations.
2. **Redundant encoding.** Domain knowledge
 may be stored across many parameter loca-
 tions at once, so erasing any single location
 leaves the pattern intact.
3. **Hyperparameter mismatch.** NPO+KL set-
 tings tuned for general-domain text may sim-
 ply be too weak for densely encoded domain
 knowledge.

The LegalBench result is consistent with the sec-
 ond explanation. Under identical hyperparameters,

Mistral-7B collapses while Saul-7B does not. Domain pretraining may stabilize the parameter space around domain-specific representations, making Saul-7B more resilient to NPO perturbation, while the base model’s parameters drift more freely under the same intervention.

5.2 What We Reproduced vs. What We Extended

Tutek et al. (2025) showed that plausibility and parametric faithfulness are weakly correlated on general-domain benchmarks. We **replicated** their pipeline on the same benchmarks using Mistral-7B-Instruct-v0.2 with the same code, segmentation strategy, NPO+KL procedure, and four evaluation metrics, confirming the methodology generalizes as reported. We **extended** the work in four ways: applying FUR to domain-adapted models (BioMistral-7B and Saul-7B) on new datasets (MedQA and LegalBench); discovering reasoning entrenchment; identifying catastrophic forgetting as a domain-specific NPO+KL failure mode; and demonstrating that unlearning hyperparameters require domain-specific tuning. Reasoning entrenchment is our contribution.

5.3 Implications

For interpretability. Domain-adapted models produce reasoning that is equally causally linked to their predictions as general-purpose models. Practitioners using CoT explanations in clinical or legal AI are not dealing with more post-hoc reasoning than in general settings, though the tools used to verify this, including FUR itself, will need domain-specific hyperparameter calibration.

For model editing. Removing knowledge, correcting bias, or making targeted fixes in domain-adapted models is substantially harder than in general-purpose ones. The same unlearning procedure either fails quietly (low efficacy, as with BioMistral) or destroys general capabilities (as with Mistral-7B on legal text). Neither outcome is acceptable in a deployment setting, pointing to the need for domain-aware regularization.

5.4 Limitations

Scale of extension experiments. Our extension results are based on 30 MedQA instances per model and 50 LegalBench instances, yielding only 31 high-efficacy steps for BioMistral. Larger samples would narrow confidence intervals and enable more precise efficacy-stratified analysis. Future work

should include larger-scale evaluations to increase statistical power and enable more fine-grained analysis of domain-specific effects.

Invalid LegalBench baseline. Without a valid Mistral-7B LegalBench run, the Saul-7B result (41.1% FF-HARD) cannot be contextualized against a reliable base model. Hyperparameter re-tuning is needed before cross-model conclusions can be drawn for the legal domain. This highlights the importance of domain-specific hyperparameter tuning for unlearning-based faithfulness measurements.

Low Saul-7B efficacy. At 20.90%, an efficacy-stratified comparison analogous to MedQA cannot be performed for LegalBench. This limits our ability to definitively separate faithfulness from modifiability in the legal domain and suggests that stronger unlearning interventions may be necessary for legal text.

6 Conclusion

We replicated the FUR pipeline from Tutek et al. (2025) on its original benchmarks and extended it to domain-adapted models on two new datasets. The main finding from the extension is that domain adaptation does not erode parametric faithfulness; it produces *reasoning entrenchment*. BioMistral’s reasoning resists unlearning in 70% of cases compared to 21% for Mistral, yet when unlearning does succeed, both models show identical FF-HARD ($\sim 67\%$, $p = 0.96$). On LegalBench, Saul-7B gives a valid faithfulness estimate (FF-HARD = 41.1%), while the Mistral-7B baseline collapses due to catastrophic forgetting, showing that standard NPO+KL settings cannot be applied to legal text without modification. Parametric faithfulness and reasoning modifiability are separable properties of domain-adapted models, and treating them as equivalent leads to incorrect conclusions about both interpretability and the feasibility of model editing. Future work should explore domain-specific regularization strategies that can successfully modify domain-adapted reasoning without causing catastrophic forgetting, and investigate whether reasoning entrenchment extends to other specialized domains beyond medicine and law.

Team Members and Roles

Madhavan Balaji led data curation, CoT segmentation, FUR implementation, and baseline metric computation.

Suman Lamsal built the unlearning pipeline (NPO+KL, layer selection, model storage) and integrated it with the evaluation framework for both MedQA and LegalBench.

Samuditha Wijayasundara designed the domain-adaptation extension, developed the statistical testing framework, led the efficacy-stratified analysis, and wrote the report.

Acknowledgments

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